



Engineering mechanics

Lecture No. 1

1-1. Introduction

1-2. Units and Dimensions, and Homogeneity

1-3. Introduction to statics

1-4. Newton's Three Laws of Motion

1-5. Scalars and vectors

*1-6. Vectors language and introduction to -
vector addition*



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Sources:

- 1- R. C. Hibbeler. Engineering mechanics statics. Fourteen edition. 2016.
- 2- R. C. Hibbeler. Engineering mechanics dynamics. Fourteen edition. 2016.

1-1. Introduction

In this course we will focus on four aspects: **Statics**, **Dynamics**.



Statics: is a branch of the physical sciences that is concerned with the state of rest or motion of bodies that are subjected to the action of forces



Dynamics: is a branch of the physical sciences that is concerned with motion of bodies

1-2: Units and Dimensions, and Homogeneity

There are two important Unit systems:

1. English System or United States Customary System (USCS): It uses in some countries. Basic units for the system are:

Dimension	Unit
Length	Foot-inch
Mass	Pound (lb)
Temperature	Fahrenheit (°F)
Time	Second (sec)

2- International System (SI) Units: It uses for most engineering fields. Basic Units are

Dimension	Unit
Length	Meter (m)
Mass	Kilogram (kg)
Temperature	Centigrade (°C)
Time	Second (sec)

Dimensional homogeneity

All equations must be dimensionally **homogeneous**.

Unity Conversion Ratios

All nonprimary units (secondary units) can be formed by combinations of primary units.

$$1 \text{ N} = 1 \text{ kg} \frac{\text{m}}{\text{s}^2} \quad \text{and} \quad 1 \text{ lbf} = 32.174 \text{ lbm} \frac{\text{ft}}{\text{s}^2}$$

EX: Spotting Errors from Unit Inconsistencies

$$E = 25 \text{ kJ} + 7 \text{ kJ/kg}$$

Multiple	Prefix	Symbol
10^{-15}	femto	f
10^{-12}	pico	p
10^{-9}	nano	n
10^{-6}	micro	μ
10^{-3}	milli	m
10^{-2}	centi	c
10^2	hecto	h
10^3	kilo	k
10^6	mega	M
10^9	giga	G
10^{12}	tera	T
10^{15}	peta	P

Quantities, dimensions and units

Quantity	Preferred units (SI)
Length (L)	m
Time (T)	s
Mass (M)	kg
Area (A)	m^2
Volume (Vol)	m^3
Velocity (V)	m/s
Acceleration (a)	m/s^2
Discharge (Q)	m^3/s
Force (F)	N
Pressure (p)	Pa
Shear stress (τ)	N/m^2
Density (ρ)	kg/m^3
Specific weight (ω)	N/m^3
Energy/Work/Heat (E)	J
Power (P)	W
Dynamic viscosity (μ)	N s/m^2 or Pa.s

Some SI and English Units

$$1 \text{ lbm} = 0.45359 \text{ kg}$$

$$1 \text{ ft} = 0.3048 \text{ m}$$

$$W = mg \quad (\text{N})$$

W weight

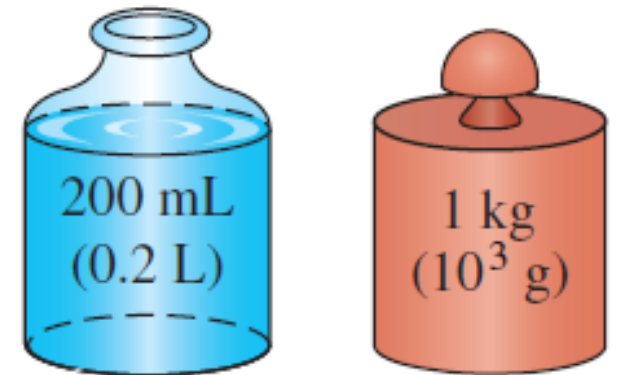
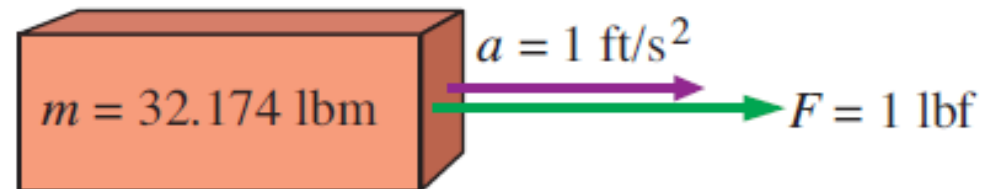
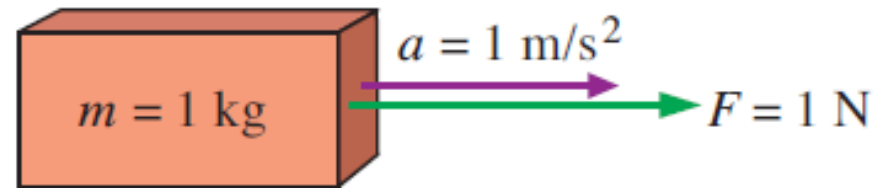
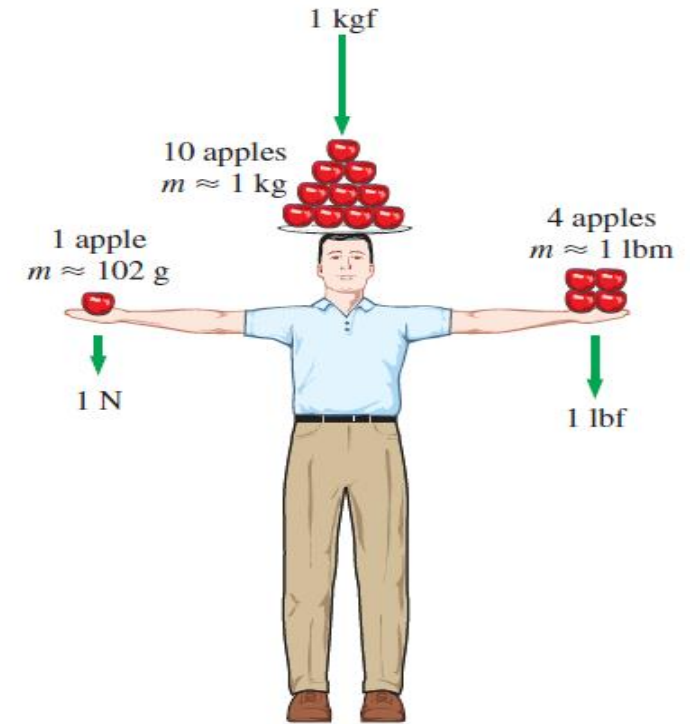
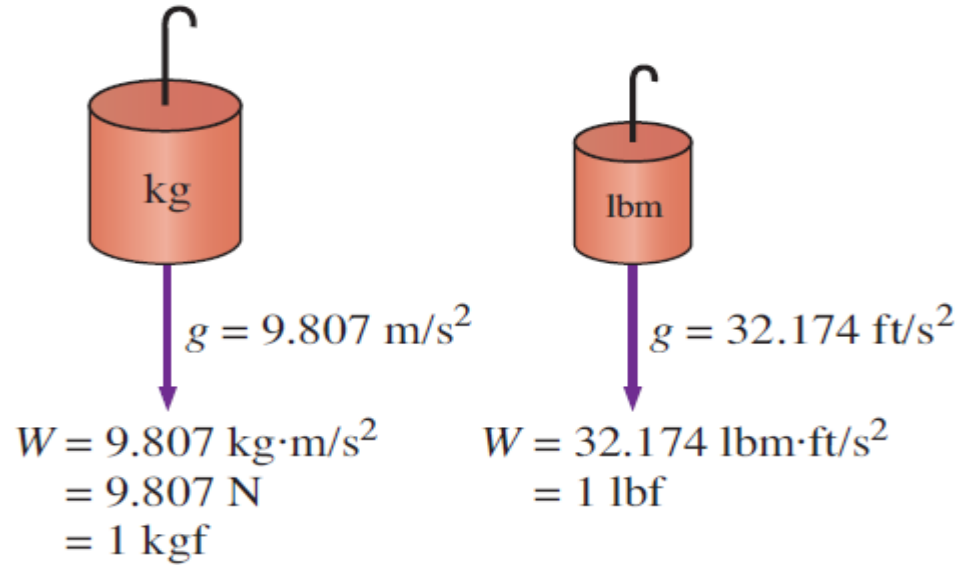
m mass

g gravitational
acceleration

$$F = ma$$

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$$

$$1 \text{ lbf} = 32.174 \text{ lbm} \cdot \text{ft/s}^2$$



1-3: Introduction to statics

Mechanics is a branch of the physical sciences that is concerned with the state of **rest or motion of bodies** that are subjected to the action of forces.

In general, this subject can be subdivided into three branches: ***rigid-body mechanics***, *deformable-body mechanics*, and *fluid mechanics*. Rigid-body mechanics is divided into two areas: **statics** and **dynamics**.

Statics deals with the equilibrium of bodies, that is, those that are either at rest or move with a constant velocity.



Basic Concepts:

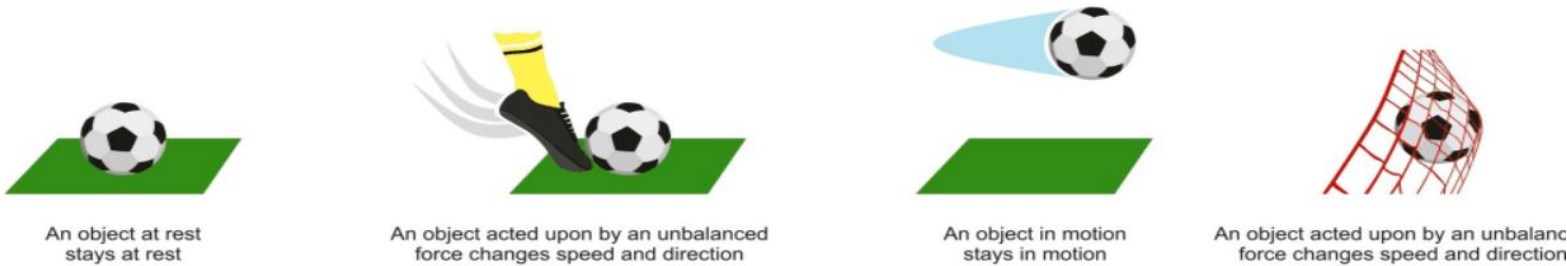
Force: is the action of one body on another. A force tends to move a body in the direction of its action. The action of a force is characterized by its **magnitude**, by the **direction** of its action, and by its point of application.

A particle : is a body of negligible dimensions. In the mathematical sense, a particle is a body whose dimensions are considered to be near zero so that we may analyze it as a mass concentrated at a point.

Rigid body: A body is considered rigid when the change in distance between any two of its points is unchangeable under and acting forces.

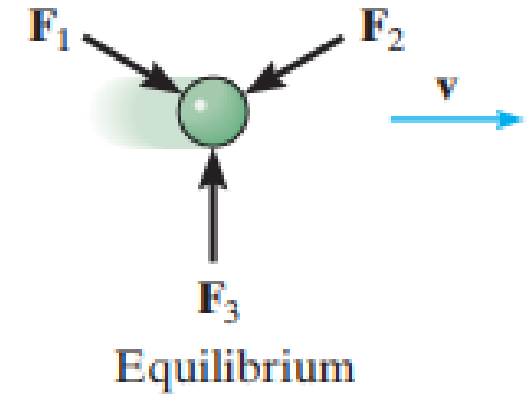
1-4: Newton's Three Laws of Motion

Law I. An object at rest will remain at rest or an object in motion will remain in motion until no unbalanced force acting on it.

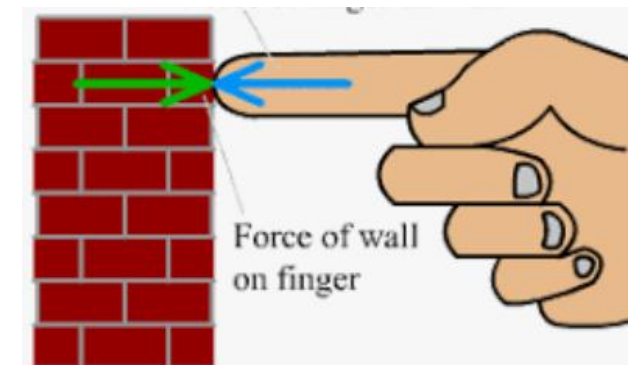


Law II: A force on a body is equal to its mass times its acceleration

Law III: For any every action, there is an equal and opposite reaction



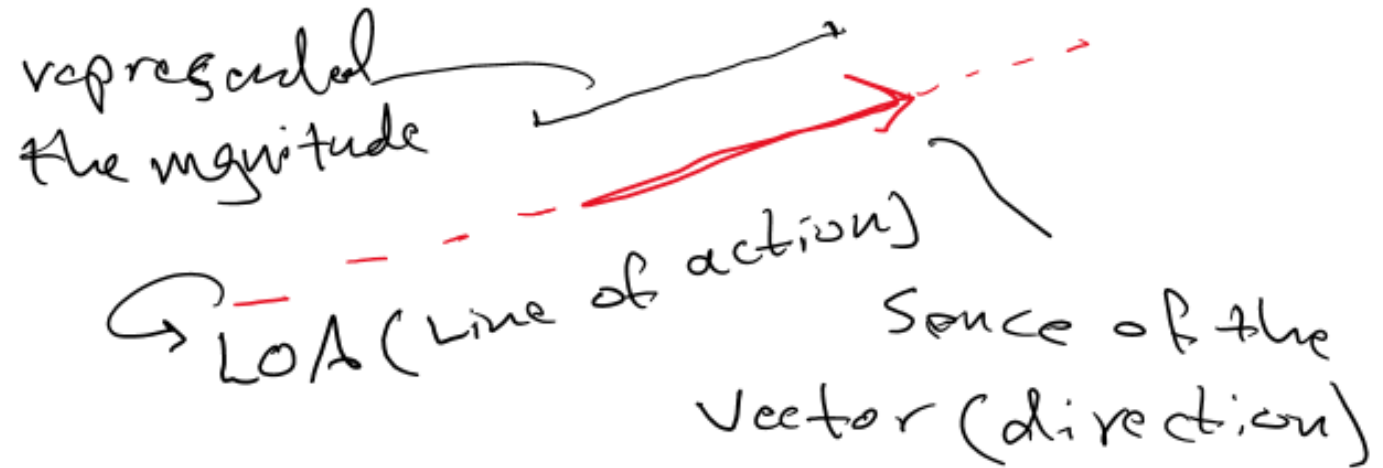
$$\mathbf{F} = m\mathbf{a}$$



1-5: Scalars and vectors

Scalars: a quantity represented by magnitude alone like mass m , temperature T , pressure p , time s , length L , and so on.

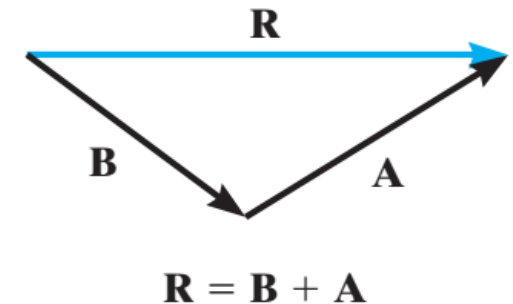
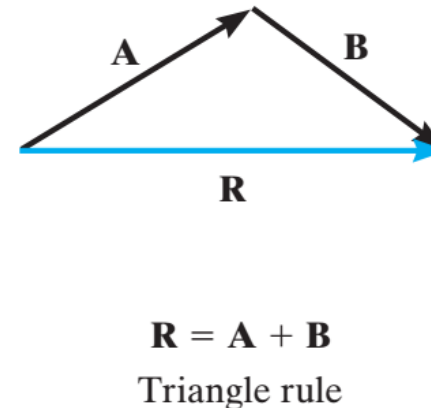
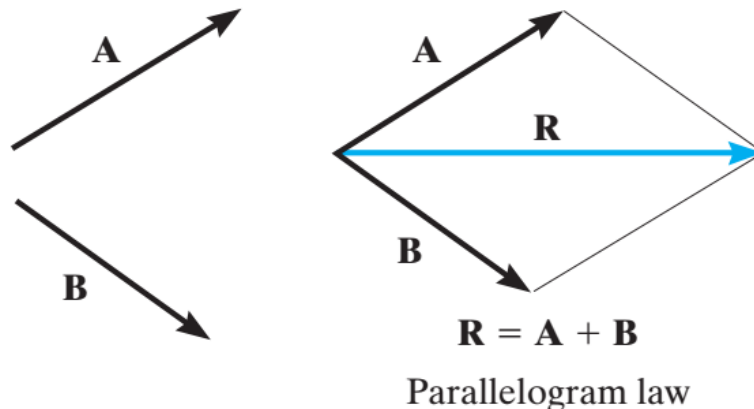
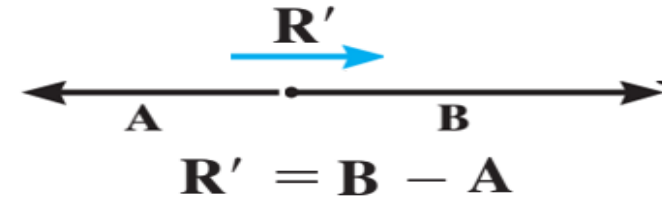
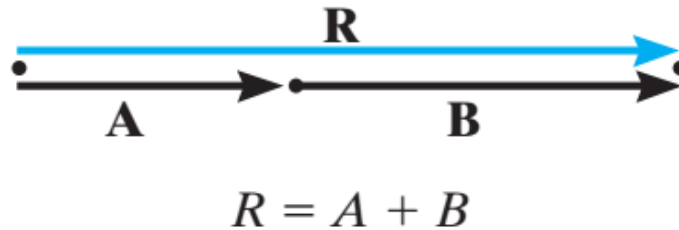
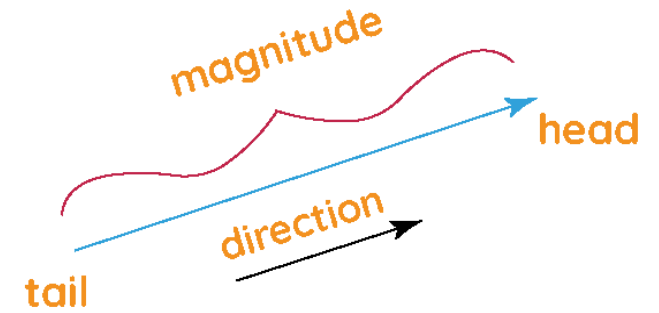
Vectors: a quantity represented by both magnitude and direction like Velocity, $\vec{V}$, Acceleration $\vec{a}$, Force $\vec{F}$, X-displacement.

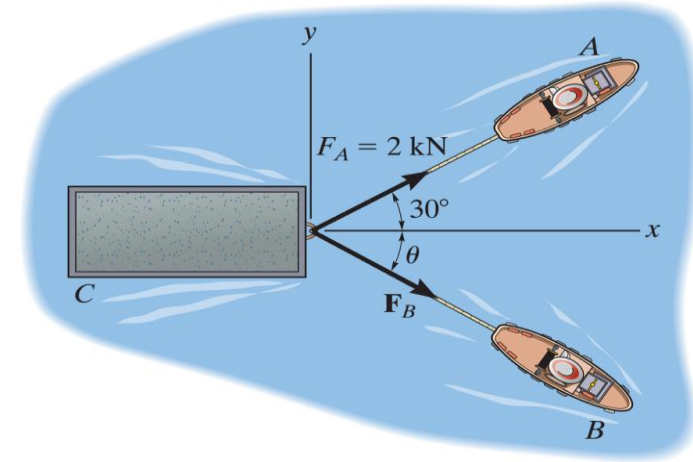
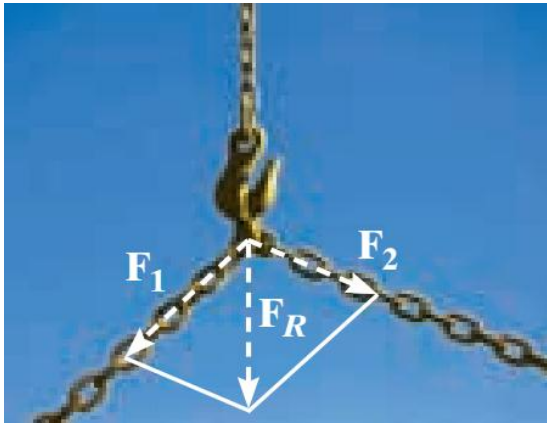


1-6: Vectors language and introduction to vector addition

Vector operations:

A vector can be represented graphically by an arrow. The length of the arrow represents the magnitude of the vector. The head of the arrow indicates the sense of direction of the vector.





➤ Trigonometry:

The sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The cosine rule

$$a^2 = b^2 + c^2 - 2bc \cos A \quad \text{and}$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

